

DiscreteEventSimulation

Discrete-event simulation in Julia: machine_shop (8 replications of 4000 minutes, seed 20260101)

Title: DiscreteEventSimulation | Model: machine_shop | Seed: 20260101 | Replications: 8 | Horizon: 4000 | Warmup: 400 |
Generated: 2026-09-24 23:23:06 | Engine: DiscreteSim | Sections: 8

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Overview

A discrete-event simulation study: the engine, the calibration and the experiments, with every number carrying its confidence interval.

MODELS 5	THROUGHPUT 0.807 items_per_minute	CYCLE TIME MEAN 14.8 minutes	UTILISATION 0.566
COMPLETED 2905	VALIDATION validated	REEVALUATION keep	

Design of the study

Seed 20260101
Days 45
Replications 8
Horizon 4000
Warmup 400
Models Any["mmc", "transfer_line", "machine_shop", "inventory", "call_center"]
Objective cycle_time_mean
Online policy offline_first

The models

Catalogue

Model	Title	Entity	Resource	Answers
mmc	ServicePool	customer	server	ServicePoolWithIdenticalServers
transfer_line	TransferLine	part	station	SerialLineWithBuffersAndFailures
machine_shop	JobShop	job	machine	JobShopWithRoutingsReworkScrapAndBreakdowns
inventory	InventoryPosition	order	stock	ReorderPointPolicyWithLeadTimeAndBackorders
call_center	ContactCenter	call	agent	AgentPoolWithPatienceAndPriority

Machine shop over 8 replications

Metric	Mean	Half width	Unit	Rel hw pct
availability	0.915	0.013	ratio	1.395

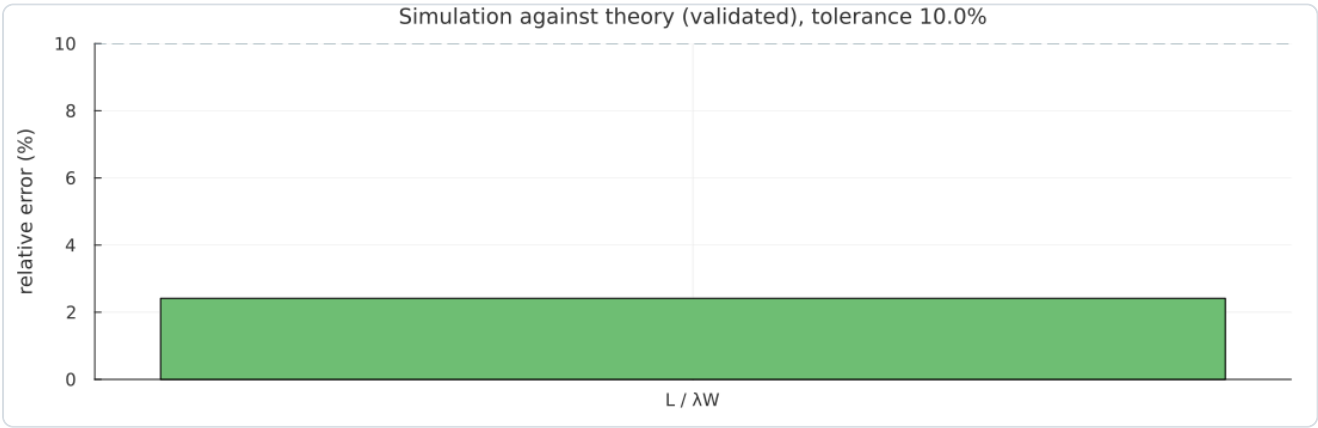
Metric	Mean	Half width	Unit	Rel hw pct
wait_p95	45.5	7.774	minutes	17.1
max_queue_length	3.522	0.968	count	27.5
queue_length_mean	9.583	1.474	count	15.4
cycle_time_mean	14.8	1.575	minutes	10.7
setup	2	0	minutes	0
scrapped	141.5	13.9	count	9.851
throughput	0.807	0.019	items_per_minute	2.398
events	27050	748.3	count	2.766
span	3600	0.122	count	0.00338
reworked	427.8	17.3	count	4.053
wait_mean	14.8	1.575	minutes	10.7
completed	2905	69.7	count	2.398
wip_mean	11.9	1.522	count	12.8
utilisation	0.566	0.013	ratio	2.219
tardiness_mean	9.855	1.548	minutes	15.7

The engine

A next-event calendar, coroutines that suspend on two channels, and statistics that know whether they are adequate.

Against theory

Verdict: validated worst relative error 2.413% against a tolerance of 10%.



Validation

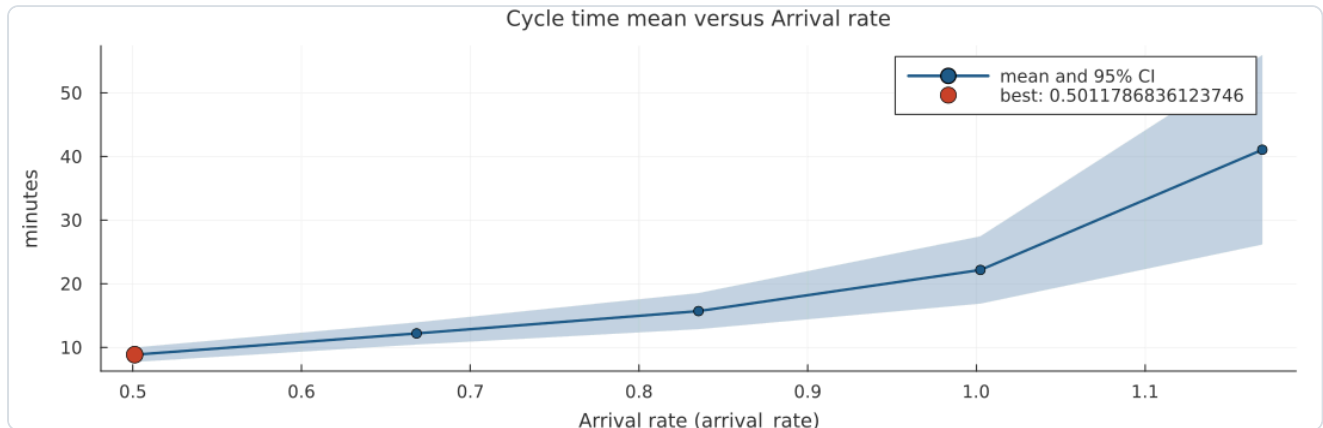
Cross checks

Observed key	Theory key	Observed	Theory	Relative error	Within tolerance
L	λW	2.098	2.149	0.024	yes

Queues and capacity

The first thing to check in a queueing model is that it agrees with Erlang, and the second is what happens when the load moves.

Widening the bottleneck



Sweep

Arrival rate

Value	Throughput mean	Wait mean	Utilisation mean	Cycle time mean
0.501	0.49	—	0.343	—
0.668	0.645	—	0.454	—
0.835	0.806	—	0.565	—
1.002	0.963	—	0.676	—
1.169	1.103	—	0.773	—

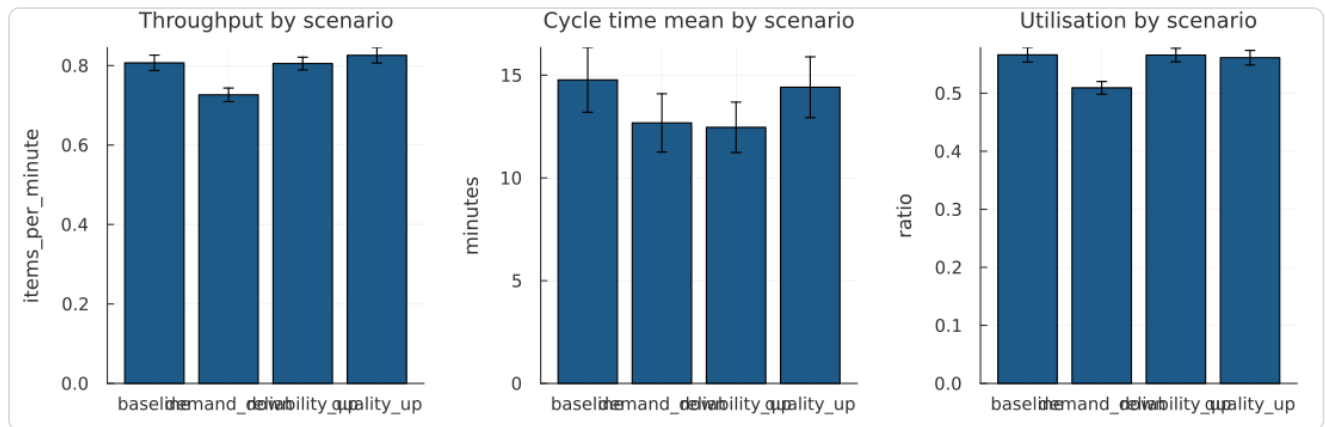
The best arrival_rate by cycle_time_mean is 0.5011786836123746 (mean 8.862 ± 1.135).

Scenarios against the baseline

Scenarios

Scenario	Metric	Baseline	Scenario mean	Difference	Half width	Relative pct	Verdict
demand_down	throughput	0.807	0.726	-0.081	0.00305	-9.99	worse
demand_down	cycle_time_mean	14.8	12.7	-2.092	1.367	-14.2	better
demand_down	wait_mean	14.8	12.7	-2.092	1.367	-14.2	better
demand_down	utilisation	0.566	0.509	-0.057	0.00231	-10	worse
demand_down	completed	2905	2614	-290.2	11	-9.992	worse
reliability_up	throughput	0.807	0.805	-0.0017	0.00397	-0.211	indistinguishable
reliability_up	cycle_time_mean	14.8	12.5	-2.309	1.195	-15.6	better
reliability_up	wait_mean	14.8	12.5	-2.309	1.195	-15.6	better
reliability_up	utilisation	0.566	0.566	-0.00036	0.00104	-0.064	indistinguishable
reliability_up	completed	2905	2898	-6.375	14.5	-0.219	indistinguishable
quality_up	throughput	0.807	0.826	0.019	0.00223	2.351	better

Scenario	Metric	Baseline	Scenario mean	Difference	Half width	Relative pct	Verdict
quality_up	cycle_time_mean	14.8	14.4	-0.357	0.157	-2.414	better
quality_up	wait_mean	14.8	14.4	-0.357	0.157	-2.414	better
quality_up	utilisation	0.566	0.561	-0.00489	0.000593	-0.863	worse
quality_up	completed	2905	2973	68.2	7.941	2.35	better



Scenarios

Best by cycle_time_mean: **reliability_up** (compared with baseline, common random numbers on every replication).

The models

Five models, one engine: a service pool, a transfer line, a job shop, an inventory position and a contact centre.

Models

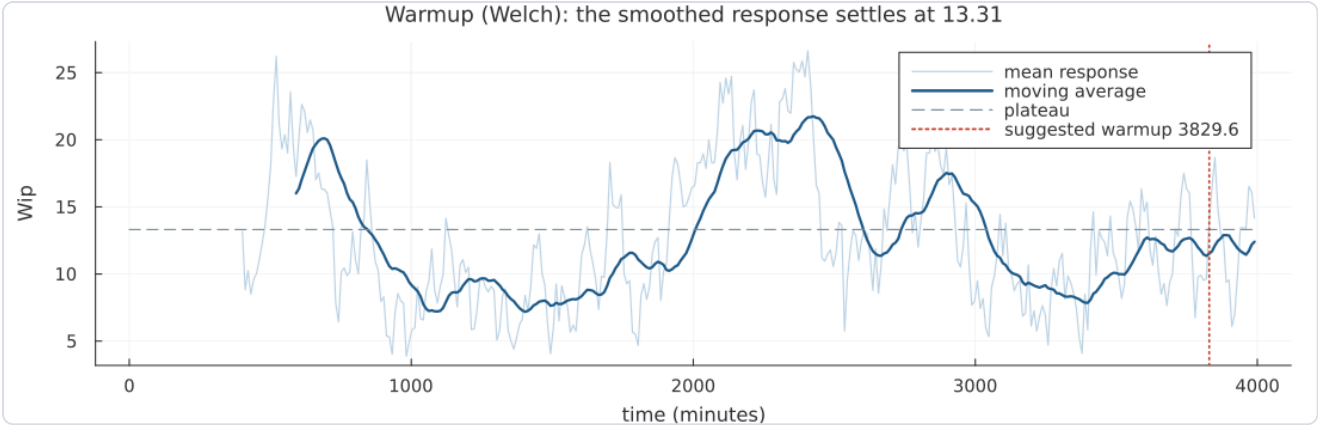
Model	Entity	Completed	Throughput	Cycle time	Wait	Utilisation	Availability
call_center	call	3041	0.845	1.341	0.011	0.281	1
machine_shop	job	2905	0.807	14.8	14.8	0.566	0.915
inventory	order	7178	1.994	—	—	—	—
transfer_line	part	685.4	0.19	96.7	96.7	0.8	0.93
mmc	customer	3048	0.847	1.912	0.582	0.563	1

Experiments

Replications, a detected warmup, batch means for a single run, paired comparisons and a factorial design -- all with common random numbers.

How long the transient lasts

Welch's procedure on the wip series over 4 replications suggests discarding the first 3830 time units (plateau 13.3 ± 1.912). That is most of the horizon: for this series the diagnostic could not find a plateau -- the curve moves by 1.912 around 13.3 all the way to the end -- so the number below is a design decision, not a measurement: more replications or a longer window would settle it.



Warmup

One long run, honestly

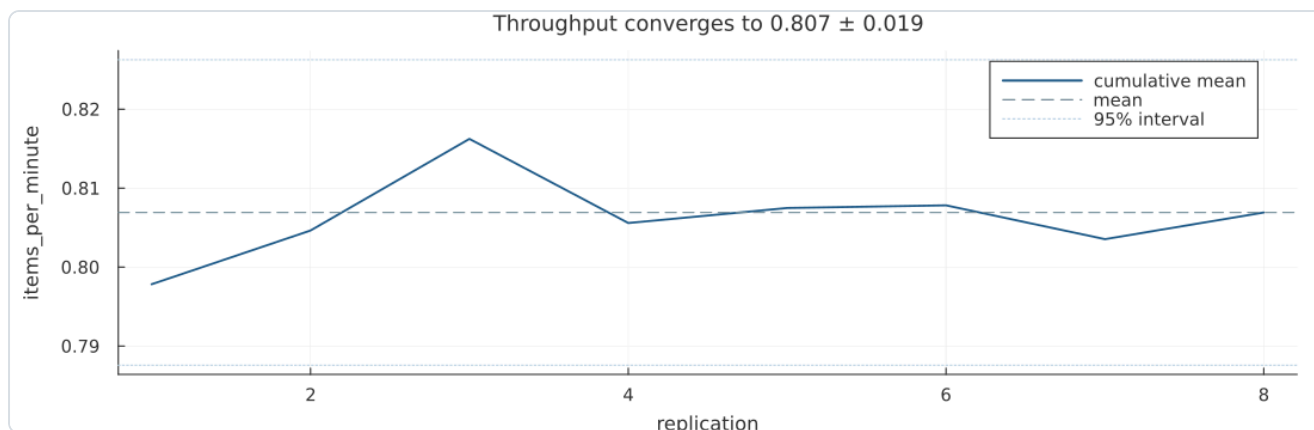
Batch means

N	Mean	Half width	Lo	Hi	Batches	Lag1	Adequate
8	14.8	2.031	12.7	16.8	4	0.219	no

The replications

Machine shop over 8 replications

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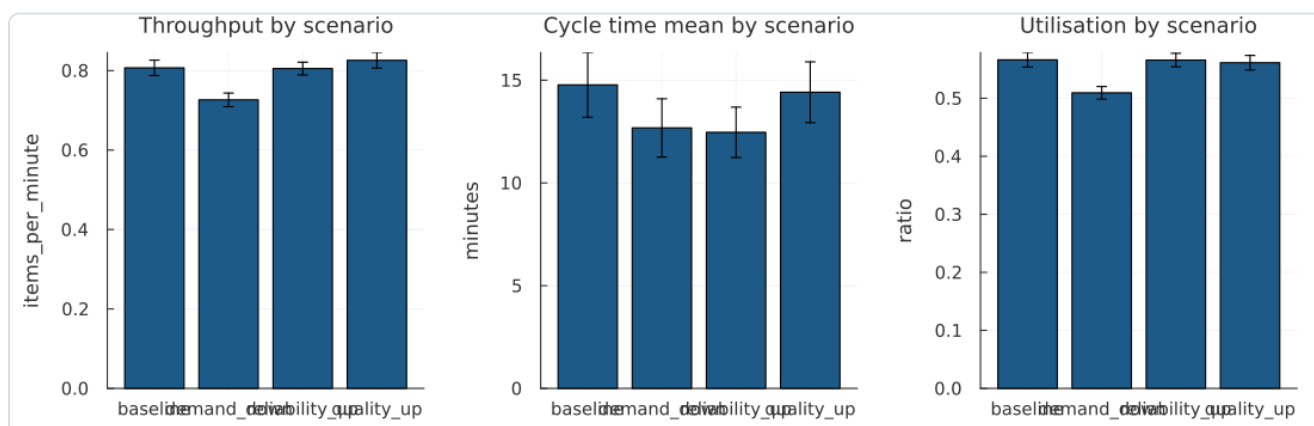


Convergence

Scenarios against the baseline

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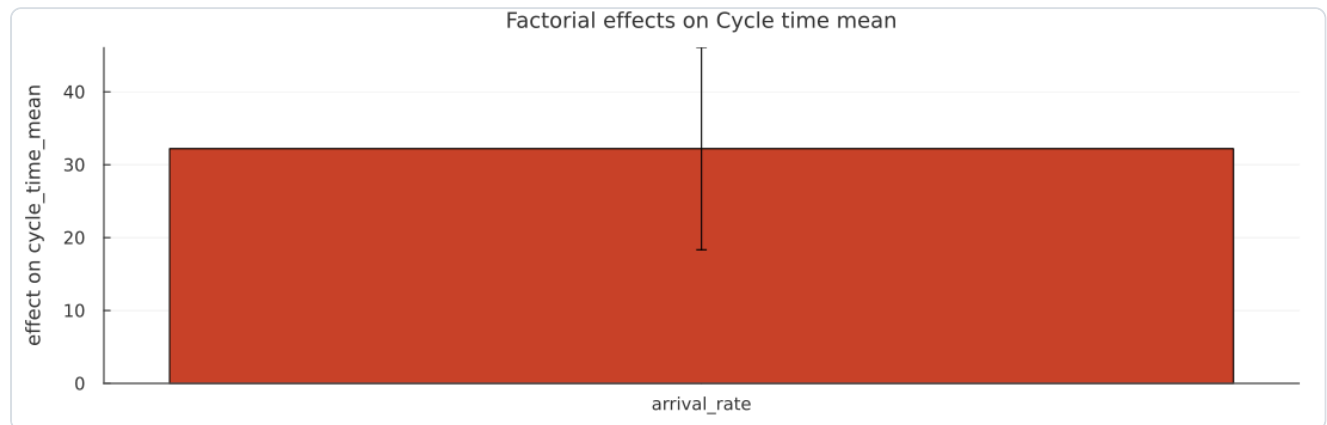
Scenarios

Best by cycle_time_mean: **reliability_up** (compared with baseline, common random numbers on every replication).

Which knob matters?

Factorial

Term	Metric	Effect	Half width	Lo	Hi	Verdict
arrival_rate	cycle_time_mean	32.2	13.9	18.3	46.1	hurts



Factorial

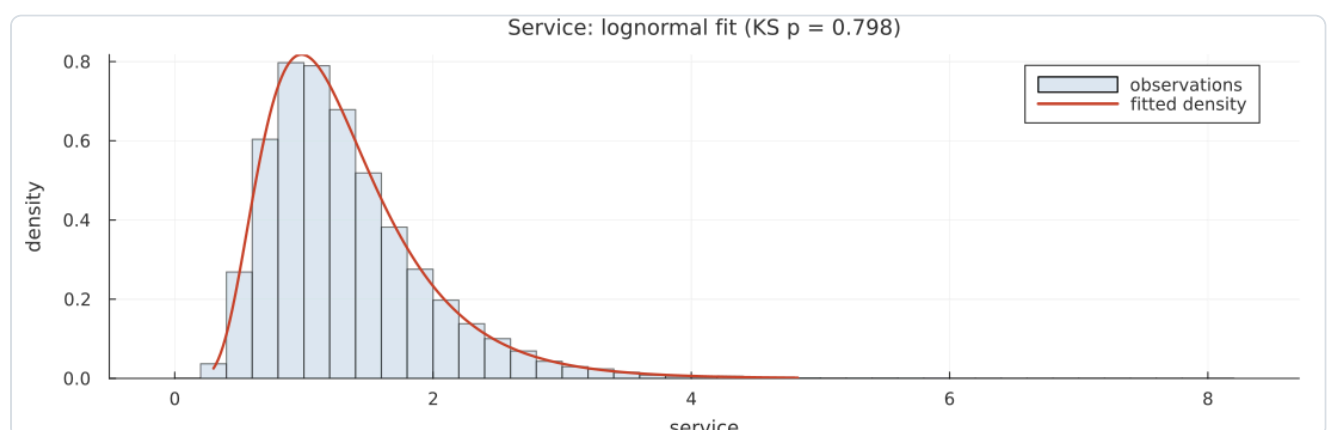
Calibration

The model parameters come from measurements, and every fit is tested against the observations it claims to describe.

Maximum-likelihood fits

Fits

Series	Family	Mean	Sample mean	Cv	N	Ks stat	Ks p	Verdict
failure_interval	weibull	172.3	172.3	0.663	354	0.024	0.988	validated
interarrival	gamma	1.197	1.197	1.005	54127	0.00293	0.741	validated
repair	lognormal	11.1	11.1	0.529	354	0.044	0.488	validated
service	lognormal	1.329	1.328	0.473	48780	0.00292	0.798	validated
demand_size	lognormal	14.3	14.4	0.488	82	0.056	0.957	validated



Calibration

What the data implies for the model

Arrival rate 0.835
Service rate 0.753
Mtbf 172.3
Mtrr 11.1
Availability 0.94
Demand mean 14.3

Offline and online

The same observations, refreshed periodically, with the decision they justify written down.

Source online status ok freshness fresh verdict keep (within_threshold)

The feed

Url https://raw.githubusercontent.com/caxelrud/Julia_Discrete_Sim_2026/main/data/online_feed.json
Policy offline_first
Status ok
Source online
Fetched at 2026-09-24T23:12:27
Age days 0
Freshness fresh
Attempts 1

What arrived

Feed observations

Series	Observations
failure_interval	354
interarrival	4000
repair	354
service	4000
demand_size	82

Offline against online

Reevaluation

Parameter	Offline	Online	Relative change	Beyond threshold
arrival_rate	0.835	0.796	-0.047	no
service_rate	0.753	0.754	0.00238	no
mtbf	172.3	172.3	0	no
mtrr	11.1	11.1	0	no
demand_mean	14.3	14.3	0	no

Are the two samples the same process?

Two sample ks

Series	Ks stat	Ks p	N offline	N online	Same process
failure_interval	0	1	354	354	yes
interarrival	0.019	0.125	54127	4000	yes
repair	0	1	354	354	yes
service	0.012	0.634	48780	4000	yes
demand_size	0	1	82	82	yes

The reevaluation log

Log

At	Source	Status	Freshness	Age days	Observations	Worst change	Verdict	Reason
2026-09-24T06:27:51	offline_cache	cached	unknown	—	8790	0.047	keep	within_threshold
2026-09-24T06:55:11	offline_cache	cached	unknown	—	8790	0.047	keep	within_threshold
2026-09-24T06:55:38	offline_cache	cached	unknown	—	8790	0.047	keep	within_threshold
2026-09-24T07:13:31	offline_cache	cached	unknown	—	8790	0.047	keep	within_threshold
2026-09-24T07:26:38	offline_cache	cached	unknown	—	8790	0.047	keep	within_threshold
2026-09-24T07:27:01	offline_cache	cached	unknown	—	8790	0.047	keep	within_threshold
2026-09-24T07:49:42	offline_cache	cached	unknown	—	8790	0.047	keep	within_threshold
2026-09-24T07:54:26	offline_cache	cached	unknown	—	8790	0.047	keep	within_threshold
2026-09-24T07:54:32	offline_cache	cached	unknown	—	8790	0.047	keep	within_threshold
2026-09-24T17:17:58	offline_cache	cached	unknown	—	8790	0.047	keep	within_threshold
2026-09-24T17:18:04	offline_cache	cached	unknown	—	8790	0.047	keep	within_threshold
2026-09-24T22:22:36	offline_cache	cached	unknown	—	8790	0.047	keep	within_threshold
2026-09-24T22:27:17	offline_cache	cached	unknown	—	8790	0.047	keep	within_threshold
2026-09-24T22:27:55	offline_cache	cached	unknown	—	8790	0.047	keep	within_threshold
2026-09-24T22:28:01	offline_cache	cached	unknown	—	8790	0.047	keep	within_threshold
2026-09-24T22:31:08	offline_cache	cached	unknown	—	8790	0.047	keep	within_threshold

At	Source	Status	Freshness	Age days	Observations	Worst change	Verdict	Reason
2026-09-24T22:32:29	offline_cache	cached	unknown	—	8790	0.047	keep	within_threshold
2026-09-24T22:32:35	offline_cache	cached	unknown	—	8790	0.047	keep	within_threshold
2026-09-24T22:37:00	offline_cache	cached	unknown	—	8790	0.047	keep	within_threshold
2026-09-24T22:39:49	offline_cache	cached	unknown	—	8790	0.047	keep	within_threshold
2026-09-24T22:42:58	offline_cache	cached	unknown	—	8790	0.047	keep	within_threshold
2026-09-24T22:46:16	offline_cache	cached	unknown	—	8790	0.047	keep	within_threshold
2026-09-24T22:53:01	offline_cache	cached	unknown	—	8790	0.047	keep	within_threshold
2026-09-24T22:57:45	offline_cache	cached	unknown	—	8790	0.047	keep	within_threshold
2026-09-24T23:01:21	online	ok	fresh	0	8790	0.047	keep	within_threshold
2026-09-24T23:03:03	online	ok	fresh	0	8790	0.047	keep	within_threshold
2026-09-24T23:03:13	online	ok	fresh	0	8790	0.047	keep	within_threshold
2026-09-24T23:12:27	online	ok	fresh	0	8790	0.047	keep	within_threshold

The model always answers from local data first; the feed only ever improves the picture, so nothing here fails when the network is down.

Artefacts

Every number, figure and PDF in this repository is regenerated by one command from one seed.

Pdf

Pdf

File	Directory
discrete_sim_report.pdf	C:\Users\inter\OneDrive_myWork\Research2026\Julia_Discrete_Sim_2026\reports\pdf
notebook_overview.pdf	C:\Users\inter\OneDrive_myWork\Research2026\Julia_Discrete_Sim_2026\reports\pdf
notebook_engine.pdf	C:\Users\inter\OneDrive_myWork\Research2026\Julia_Discrete_Sim_2026\reports\pdf
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notebook_experiments.pdf	C:\Users\inter\OneDrive_myWork\Research2026\Julia_Discrete_Sim_2026\reports\pdf
notebook_calibration.pdf	C:\Users\inter\OneDrive_myWork\Research2026\Julia_Discrete_Sim_2026\reports\pdf

File	Directory
notebook_online.pdf	C:\Users\inter\OneDrive\myWork\Research2026\Julia_Discrete_Sim_2026\reports\pdf
notebook_reports.pdf	C:\Users\inter\OneDrive\myWork\Research2026\Julia_Discrete_Sim_2026\reports\pdf

Data

Data

File	Directory
observations_interarrival.csv	C:\Users\inter\OneDrive\myWork\Research2026\Julia_Discrete_Sim_2026\data
observations_service.csv	C:\Users\inter\OneDrive\myWork\Research2026\Julia_Discrete_Sim_2026\data
observations_failure_interval.csv	C:\Users\inter\OneDrive\myWork\Research2026\Julia_Discrete_Sim_2026\data
observations_repair.csv	C:\Users\inter\OneDrive\myWork\Research2026\Julia_Discrete_Sim_2026\data
observations_demand_size.csv	C:\Users\inter\OneDrive\myWork\Research2026\Julia_Discrete_Sim_2026\data
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online_feed.json	C:\Users\inter\OneDrive\myWork\Research2026\Julia_Discrete_Sim_2026\data

Figures

Figures

File	Directory
wip.png	C:\Users\inter\OneDrive\myWork\Research2026\Julia_Discrete_Sim_2026\reports\figures
wait.png	C:\Users\inter\OneDrive\myWork\Research2026\Julia_Discrete_Sim_2026\reports\figures
utilisation.png	C:\Users\inter\OneDrive\myWork\Research2026\Julia_Discrete_Sim_2026\reports\figures
throughput.png	C:\Users\inter\OneDrive\myWork\Research2026\Julia_Discrete_Sim_2026\reports\figures
gantt.png	C:\Users\inter\OneDrive\myWork\Research2026\Julia_Discrete_Sim_2026\reports\figures
scenarios.png	C:\Users\inter\OneDrive\myWork\Research2026\Julia_Discrete_Sim_2026\reports\figures
sweep.png	C:\Users\inter\OneDrive\myWork\Research2026\Julia_Discrete_Sim_2026\reports\figures
warmup.png	C:\Users\inter\OneDrive\myWork\Research2026\Julia_Discrete_Sim_2026\reports\figures
validation.png	C:\Users\inter\OneDrive\myWork\Research2026\Julia_Discrete_Sim_2026\reports\figures
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calibration.png	C:\Users\inter\OneDrive\myWork\Research2026\Julia_Discrete_Sim_2026\reports\figures
convergence.png	C:\Users\inter\OneDrive\myWork\Research2026\Julia_Discrete_Sim_2026\reports\figures

Html

Html

File	Directory
discrete_sim_report.html	C:\Users\inter\OneDrive\myWork\Research2026\Julia_Discrete_Sim_2026\reports\html
notebook_overview.html	C:\Users\inter\OneDrive\myWork\Research2026\Julia_Discrete_Sim_2026\reports\html
notebook_engine.html	C:\Users\inter\OneDrive\myWork\Research2026\Julia_Discrete_Sim_2026\reports\html

File	Directory
notebook_queues.html	C:\Users\inter\OneDrive_myWork\Research2026\Julia_Discrete_Sim_2026\reports\html
notebook_models.html	C:\Users\inter\OneDrive_myWork\Research2026\Julia_Discrete_Sim_2026\reports\html
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Reproduce everything with one command and one seed: `julia --project=. scripts/run_study.jl --seed=20260101`.

Generated by DiscreteSim.jl from seed 20260101 on 2026-09-24 23:23:06; every number is reproducible.